2018-2019 微积分 1-2 半期试题参考解答

1.(10分) 求极限 $\lim\limits_{\substack{x\to 0\\ y\to 2}}\left(\dfrac{\sin xy}{xy}\right)^{\frac{1}{x^2}}$.

解：原式 $=\lim\limits_{\substack{x\to 0\\ y\to 2}} e^{\frac{\ln\frac{\sin xy}{xy}}{x^2}}=\lim\limits_{\substack{x\to 0\\ y\to 2}} e^{\frac{\sin xy-xy}{x^3y}}=\lim\limits_{\substack{x\to 0\\ y\to 2}} e^{\frac{-\frac{1}{6}(xy)^3}{x^3y}}=e^{-\frac{2}{3}}$.

2.(10分)设函数 $u=f(z)$**可微，方程** $z=y+x\varphi(z)$**可确定** z **是** x,y **的函数，** $\varphi(z)$**可微，试求** $\dfrac{\partial u}{\partial x}-\varphi(z)\dfrac{\partial u}{\partial y}$.

解：令 $F(x,y,z)=y+x\varphi(z)-z$**，则** $F_x'=\varphi(z),F_y'=1,F_z'=\varphi'(z)-1$**. 故**

$$\frac{\partial z}{\partial x}=-\frac{F_x'}{F_z'}=-\frac{\varphi(z)}{\varphi'(z)-1},\frac{\partial z}{\partial y}=-\frac{F_y'}{F_z'}=-\frac{1}{\varphi'(z)-1},$$

所以 $\dfrac{\partial u}{\partial x}-\varphi(z)\dfrac{\partial u}{\partial y}=f'(z)\cdot\dfrac{\partial z}{\partial x}-\varphi(z)f'(z)\cdot\dfrac{\partial z}{\partial y}=0.$

3.(12分)设 $f(x,y)=\begin{cases}xy\arctan\dfrac{1}{\sqrt{x^2+y^2}},&(x,y)\neq(0,0)\\ 0,&(x,y)=(0,0)\end{cases}$**，讨论在点** $(0,0)$**处的连续性、偏导数的存在性及可微性.**

解：(1)因为 $0\le|f(x,y)|\le\dfrac{\pi}{2}|xy|\to 0,(x,y)\to(0,0)$**，所以** $\lim\limits_{(x,y)\to(0,0)}f(x,y)=0$**，连续.**

(2) $f_x'(0,0)=\lim\limits_{(x,y)\to(0,0)}\dfrac{f(x,0)-f(0,0)}{x}=0$**，同理** $f_y'(0,0)=0.$

(3)因为 $\lim\limits_{(x,y)\to(0,0)}\dfrac{f(x,y)-f_x'(0,0)x-f_y'(0,0)y}{\sqrt{x^2+y^2}}=\lim\limits_{(x,y)\to(0,0)}\dfrac{xy\arctan\frac{1}{\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}}$**.而**

$$0\le\left|\frac{xy\arctan\frac{1}{\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}}\right|\le\frac{\pi}{2}\frac{|x|}{\sqrt{x^2+y^2}}|y|\le\frac{\pi}{2}|y|\to 0,(x,y)\to(0,0).$$

所以可微.

4.(10分)求曲面 Σ**：** $x^2+2y^2+z^2=13$ **到平面** π**:** $2x+4y+z=20$**的最短距离.**

解：曲面上任意点 $M(x,y,z)$**到平面的距离为** $d=\dfrac{|2x+4y+z-20|}{\sqrt{21}}$.

作拉格朗日函数 $L(x,y,z,\lambda)=(2x+4y+z-20)^2+\lambda(x^2+2y^2+z^2-13).$

令$\begin{cases} L'_x = 4(2x+4y+z-20)+2\lambda x = 0 \\ L'_y = 8(2x+4y+z-20)+4\lambda y = 0 \\ L'_z = 2(2x+4y+z-20)+2\lambda z = 0 \\ L'_\lambda = x^2+2y^2+z^2-13 = 0 \end{cases}$，解之得$\begin{cases} x=2 \\ y=2 \\ z=1 \end{cases}$或$\begin{cases} x=-2 \\ y=-2 \\ z=-1 \end{cases}$.

当$x=2, y=2, z=1$时，$d=\frac{7}{\sqrt{21}}$；当$x=-2, y=-2, z=-1$时，$d=\frac{33}{\sqrt{21}}$，故最短距离为$\frac{7}{\sqrt{21}}$.

5.(12分)计算$I=\int_L (e^x \sin y - y - y^3)dx + (e^x \cos y + x)dy$，其中$L$为从点$O(0,0)$沿$y=\sqrt{2x-x^2}$至点$A(2,0)$的上半圆周.

解：作辅助有向曲线$\overline{AO}$，设L与$\overline{AO}$围成的区域为D，令

$$P = e^x \sin y - y - y^3, Q = e^x \cos y + x,$$

则$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2 + 3y^2$.由格林公式

$$\text{原式} = \oint_{L+\overline{AO}} - \int_{\overline{AO}} = -\iint_D (2+3y^2)dxdy + 0$$

$$\text{原式} = \oint_{L+\overline{AO}} - \int_{\overline{AO}} = -\iint_D (2+3y^2)dxdy + 0 = -\pi - \int_0^{\frac{\pi}{2}} d\theta \int_0^{2\cos\theta} r^2 \sin^2\theta r dr = -\frac{11}{8}\pi.$$

6.(10分)计算$I=\iint_\Sigma (x+y+z)^2 dS$，其中$\Sigma$为曲面$z=\sqrt{x^2+y^2}$介于$z=0$与$z=1$之间的部分.

解：Σ在Oxy面上的投影区域$D_{xy}: x^2+y^2 \le 1$，而$\sqrt{1+(z'_x)^2+(z'_y)^2} = \sqrt{2}$，由对称性，

$$I = \iint_\Sigma (x^2+y^2+z^2)dS = \iint_{D_{xy}} 2(x^2+y^2)\sqrt{2}dxdy$$

$$= 2\sqrt{2}\int_0^{2\pi} d\theta \int_0^1 r^2 r dr = \sqrt{2}\pi.$$

7.(12分)计算$I=\iint\limits_\Sigma (x+1)\,dy\,dz + (x^2y+z^4)\,dz\,dx + (xy^3+y^2z)\,dx\,dy$，其中有向曲面$\Sigma$为$z=x^2+y^2 (0 \le z \le 1)$的上侧.

解：添加辅助有向曲面$\Sigma_1: z=1, (x,y) \in D_{xy}: x^2+y^2 \le 1$，取下侧. 设$\Sigma$与$\Sigma_1$所围成的闭区域为$\Omega$，其中

$P = x+1, Q = x^2y+z^4, R = xy^3+y^2z, \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 1 + x^2 + y^2$. 由高斯公式，

$$I = \oiint_{\Sigma+\Sigma_1} - \iint_{\Sigma_1} = -\iiint_\Omega (1+x^2+y^2)dv - \iint_{\Sigma_1} (xy^3+y^2z)\,dx\,dy$$

$$= -\iint_{D_{xy}} dxdy \int_{x^2+y^2}^{1} (1+x^2+y^2)dz + \iint_{D_{xy}} y^2 dxdy$$

$$= -\iint_{D_{xy}} (1-(x^2+y^2)^2)dxdy + \frac{1}{2}\iint_{D_{xy}} (x^2+y^2)dxdy$$

$$= -\int_0^{2\pi} d\theta \int_0^1 (1-r^4) r dr + \frac{1}{2}\int_0^{2\pi} d\theta \int_0^1 r^2 r dr$$

$$= -\frac{5}{12}\pi.$$

8.(12分)求曲线积分 $I=\int_L (1+z)dx+(2+x)dy+(3+y)dz$，其中 $L:\begin{cases} x^2+y^2=1 \\ x+y+z=3 \end{cases}$，从 z 轴正向看去取逆时针方向.

解：设L所围成的平面区域为Σ，取上侧，则其上任一点的单位法矢量

$$n^0 = (\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}).$$

由斯托克斯公式，

$$I = \iint_{\Sigma} \begin{vmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 1+z & 2+x & 3+y \end{vmatrix} dS = \frac{1}{\sqrt{3}}\iint_{\Sigma} 3\,dS = \frac{3}{\sqrt{3}}\iint_{D_{xy}} \sqrt{3}\,dxdy = 3\pi.$$

9.(12分)设函数 $\varphi(x,y)$ 有连续偏导数，曲线积分 $\int_L 2xy\,dx+\varphi(x,y)dy$ 与路径无关，且对任意 t，有 $\int_{(0,0)}^{(t,1)} 2xy\,dx+\varphi(x,y)dy = \int_{(0,0)}^{(1,t)} 2xy\,dx+\varphi(x,y)dy$，求 $\varphi(x,y)$.

解：由题意，有 $\frac{\partial \varphi(x,y)}{\partial x} = \frac{\partial (2xy)}{\partial y} = 2x$，所以 $\varphi(x,y) = x^2 + g(y)$. 同时

$$\int_{(0,0)}^{(t,1)} 2xy\,dx+\varphi(x,y)dy = \int_{(0,0)}^{(t,0)} 2xy\,dx + \int_{(t,0)}^{(t,1)} (x^2+g(y))dy = 0 + \int_0^1 (t^2+g(y))dy = t^2 + \int_0^1 g(y)dy.$$

$$\int_{(0,0)}^{(1,t)} 2xy\,dx+\varphi(x,y)dy = \int_{(0,0)}^{(1,0)} 2xy\,dx + \int_{(1,0)}^{(1,t)} (x^2+g(y))dy = \int_0^t (1+g(y))dy\ .$$

故 $\int_0^t (1+g(y))dy = t^2 + \int_0^1 g(y)dy$，两边同时对 t 求导，得 $1+g(t)=2t$，所以 $g(y)=2y-1$，因此

$$\varphi(x,y) = x^2 + 2y - 1.$$